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GCA-ODN: A global context-aware dropout network for joint facial landmark detection and emotion recognition under occlusion

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ABSTRACT

We propose the Global Context-Aware Dropout Network (GCA-ODN), a CNN-based, computationally practical neural architecture for *joint* facial landmark detection (FLD) and facial expression recognition (FER) under partial facial occlusion. GCA-ODN learns a shared embedding that encodes facial geometry and affective cues, improving reliability under face masks, large pose variations, and non-uniform illumination.

The model integrates three complementary components: (i) a global context module that captures long-range spatial dependencies and injects holistic structural cues into local features, (ii) an attentive dropout module that generates attention-guided masks to suppress unreliable or overly dominant activations and reduce dependence on a limited set of visible cues, and (iii) a low-rank feature refinement module that promotes compact, noise-suppressed representations.

Experiments across multiple FLD and FER benchmarks demonstrate competitive performance under the stated evaluation protocols, with clear improvements over the controlled ResNet-18 baseline and several occlusion-aware CNN methods. On 300W, GCA-ODN achieves NRMSE of 2.60×10^{-2} (Common), 3.20×10^{-2} (Full), and 5.61×10^{-2} (Challenging). For cross-dataset FER trained on AffectNet and evaluated directly on the target datasets without target-domain fine-tuning, it reaches 84.6% on CK+, 62.1% on JAFFE, and 68.3% on ISED. At an input resolution of 224×224 , the model processes more than 80 frames per second with batch size one on an NVIDIA RTX A6000, demonstrating real-time throughput under the reported hardware configuration.

1. Introduction

Facial analysis under occlusion remains a central challenge for neural perception systems deployed in unconstrained environments. In applications such as telemedicine, affect-aware human-computer interaction, social robotics, and mobile sensing, faces are frequently captured under *partial observation* due to masks, hands, hair, accessories, large pose variations, and non-uniform illumination (Fu et al., 2023; Liedtke et al., 2018; Onishi, 2021; Yildirim-Celik et al., 2022). These factors remove or corrupt informative evidence and can lead to unstable predictions, particularly when models overfit to a small set of salient visible cues. Two tasks are especially sensitive: facial landmark detection (FLD), which localizes facial key points, and facial expression recognition (FER), which classifies affective states from facial appearance and geometry.

Facial landmarks provide a compact geometric representation and are foundational for downstream tasks including face alignment, action unit analysis, and 3D face modeling. Traditional FLD methods include regression-based approaches (Wu & Ji, 2019), template-based models with statistical priors (e.g., PCA) (Asthana et al., 2013; Zhu & Ramanan, 2012), and deep learning (DL) methods. Although convolu-

tional neural networks (CNNs) have substantially improved FLD accuracy (Zeiler & Fergus, 2014), robustness under occlusion remains difficult because corrupted regions can dominate local feature learning and long-range geometric dependencies are not explicitly preserved. Similarly, many FER systems rely strongly on appearance cues that are vulnerable to partial visibility and distribution shifts, making cross-dataset generalization challenging in the presence of occlusion and pose changes.

Recent transformer-based FLD and FER methods model long-range facial relationships through prototype-aware, muscle-movement-aware, or texture-affinity-aware representations (Hu et al., 2026; Liu et al., 2026a, 2024a). These developments demonstrate the value of global dependency modeling, but they do not remove the need for accurate local spatial representations or deployment-conscious computation. For the present joint task, we retain a compact CNN backbone because its locality and translation-related inductive biases are well suited to dense landmark localization. The proposed Global Context Module complements this local representation by explicitly aggregating long-range facial information. The design therefore seeks a controlled accuracy-throughput balance rather than assuming that CNNs are universally superior to transformer architectures.

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Two recent studies are particularly relevant to the present problem formulation. CAD-Net combines recurrent criss-cross geometry modeling, attentive dropout, and a factorized low-rank regression head for occlusion-robust facial analysis (Sadiq et al., 2026b). A separate occlusion-aware FER framework integrates an explicit occlusion detector, image masking, low-rank semantic feature recovery, and non-occluded-image guidance (Wang et al., 2026). These approaches share the broader motivation of suppressing occlusion-corrupted evidence while preserving facial structure, but their architectures, optimization procedures, datasets, and experimental measurements are distinct from those of GCA-ODN. GCA-ODN is formulated as a shared FLD-FER representation-learning framework and does not use the external occlusion-detection and paired clean-occluded guidance pipeline of the FER-only framework. Detailed methodological distinctions are provided in Section 2.

A common limitation in the literature is that FLD and FER are often optimized as separate pipelines, even though expression cues arise from coordinated geometric changes and appearance variations. Joint modeling can therefore be beneficial: geometry supervision can regularize representation learning for FER, while expression-related structure can provide complementary cues for landmark localization. The key challenge is to design a computationally practical neural architecture that (i) preserves global facial structure, (ii) suppresses corrupted evidence during learning, and (iii) yields compact representations stable under partial observation.

To address this, we propose the **Global Context-Aware Dropout Network (GCA-ODN)**, an end-to-end architecture for *joint* FLD and FER under occlusion. GCA-ODN couples three mechanisms in a single workflow: (1) global context aggregation to capture long-range dependencies and preserve holistic geometric consistency; (2) attention-guided dropout to suppress unreliable activations and prevent over-reliance on dominant visible cues; and (3) low-rank feature refinement to reduce redundancy and noise, producing compact shared embeddings for both landmark localization and expression classification. The three components are optimized jointly so that context aggregation operates on reliability-aware features and low-rank refinement constrains the resulting shared representation. This coordinated formulation is intended to address occlusion-induced structural instability rather than to obtain improvements solely through backbone scaling. To ensure fair assessment, we report results for GCA-ODN trained on 300W as well as a variant incorporating additional Menpo training data, denoted as **GCA-ODN(M)** (Section 5).

1.1. Main contributions

The main contributions of this work are:

1. An end-to-end joint FLD-FER framework that learns a shared representation under occlusion, enabling information transfer between facial geometry and affective cues.
2. A tightly coupled tri-module design integrating global context modeling, attention-guided dropout masking, and low-rank feature refinement to improve robustness to missing or corrupted facial evidence while processing more than 80 frames per second at the stated input resolution and RTX A6000 evaluation setting.
3. Evaluation on multiple FLD and FER benchmarks covering relatively controlled images, severe occlusion, large pose variations, video sequences, and cross-dataset FER transfer, together with leave-one-module-out analysis and consolidated benchmark summaries (Tables 7–11, and 13). The evaluation distinguishes the base model from the Menpo-assisted variant and separates directly comparable results from literature values obtained using different training data or protocols. On 300W, GCA-ODN achieves NRMSE of 2.60×10^{-2} (Common), 3.20×10^{-2} (Full), and 5.61×10^{-2} (Challenging) (Tables 8 and 9); it also achieves >80 FPS on 224×224 inputs on an RTX A6000 (Table 6).

The remainder of this paper is organized as follows. Section 2 reviews related background and context. Section 3 presents the proposed GCA-ODN architecture, and Section 4 describes end-to-end optimization. Section 5 reports FLD experiments, while Section 6 reports FER experiments. Section 7 provides ablation analysis, Section 8 discusses applications, and Section 9 concludes the paper.

2. Related background and context

2.1. Facial landmark detection under occlusion

Occlusion-robust facial analysis has evolved from explicit geometric constraints and local appearance modeling toward attention-based and relationship-aware neural representations. For facial landmark detection (FLD), the goal is to localize a set of predefined facial key points, but performance degrades substantially under *partial observation* when facial regions are covered by masks, hands, hair, accessories, or medical devices (Fig. 1,2). Since occlusions are often irregular in shape, scale, and location, models must distinguish reliable evidence from corrupted cues and avoid error propagation. Early occlusion-aware FLD methods incorporate explicit visibility reasoning and robust regression, including supervised regression for updating landmark visibility probabilities (Wu & Ji, 2019), occlusion dictionaries (Xing et al., 2017), and adaptive regression with shape modeling for occlusion assessment (Liu et al., 2016). More recent approaches improve robustness through stronger detectors, better alignment strategies, and refined regression pipelines (Deng et al., 2020; Sadiq et al., 2022). Despite these advances, occlusion remains challenging because local corruption can dominate feature learning and long-range geometric dependencies are not always preserved when informative regions are missing.

Heatmap-based FLD subsequently improved localization by combining dense spatial prediction with higher-order geometric information. The Multi-order High-precision Hourglass Network combines sub-pixel heatmap regression with cross-order geometric modeling to address heavy occlusion and large pose changes (Wan et al., 2021). The Multi-order Multi-constraint Deep Network further models higher-order spatial and channel relationships while imposing global shape constraints on predicted landmarks (Wan et al., 2022). Transformer-based approaches provide another mechanism for modeling relations among spatially separated landmarks. ProtoFormer, for example, employs an adaptive prototype-aware encoder and a progressive prototype-aware decoder to support joint learning across datasets with heterogeneous landmark definitions (Hu et al., 2026). These methods demonstrate the value of heatmap localization, global structure, and relationship modeling, but their principal objective is FLD rather than learning a shared representation jointly supervised by FLD and FER.

Recent occlusion-adaptive CNN frameworks also combine feature suppression with structural regularization. ADODN employs attention-guided dropout to reduce dependence on occlusion-corrupted responses (Sadiq et al., 2026a). CAD-Net combines recurrent criss-cross geometry modeling, an attentive dropout block, and a factorized low-rank regression head within a separate occlusion-adaptive facial-analysis architecture (Sadiq et al., 2026b). Although CAD-Net and GCA-ODN share the broad motivation of preserving facial structure while suppressing unreliable evidence, they use different context operators and different placements of low-rank modeling. CAD-Net applies recurrent directional criss-cross attention and constrains its landmark regression mapping, whereas GCA-ODN employs global context aggregation and low-rank refinement of the shared representation used by its task heads. Their experimental results, efficiency measurements, and hyperparameter studies are therefore treated as model-specific rather than interchangeable.

2.2. Facial expression recognition and geometric context

Facial expression recognition (FER) is closely related to FLD because expressions manifest through coordinated changes in facial



Fig. 1. Representative occlusions in the COFW dataset caused by hair, hands, clothing, accessories, and other objects. These examples illustrate the loss of local facial evidence under partial observation and motivate the need to preserve long-range facial structure.

Table 1

Illustrative FACS/EMFACS configurations for basic emotions and their corresponding regions in a 68-point landmark layout, based on the AU definitions in FACS and EMFACS (Ekman & Friesen, 1978; Hager et al., 2002). The landmark indices provide an implementation-oriented geometric interpretation and are not an official component of the FACS specification.

Emotion	AUs	Descriptions	Facial Landmarks
Happiness	6, 12	Cheek raiser, lip corner puller	1–2, 14–15; 48–49, 53–55, 59–60, 64
Anger	4, 5 + 7, 23	Brow lowerer, upper lid raiser + lid tightener, lip tightener	17–22; 37–39, 42–44; 48–49, 53–67
Fear	1, 2, 4, 5 + 7, 20, 26	Brow actions, lid actions, lip stretcher, jaw drop	17–26; 37–39, 42–44; 48–49, 53–55, 59–60, 64; 55–66
Sadness	1, 4, 15	Inner brow raiser, brow lowerer, lip corner depressor	17–21; 22–26; 48–49, 53–55, 59–60, 64
Disgust	9, 15, 16	Nose wrinkler, lip corner depressor, lower lip depressor	27–35; 48–49, 53–58, 59–60, 64
Surprise	1, 2, 5, 26	Brow raiser, upper lid raiser, jaw drop	17–26; 37–39, 42–44; 55–66

geometry and appearance. A classical framework for describing facial movements is the Facial Action Coding System (FACS) (Ekman & Friesen, 1978), which decomposes facial behavior into action units (AUs) corresponding to anatomically grounded muscle activations. FACS provides a standardized vocabulary that motivates interpretable AU-based learning, but manual AU coding is labor-intensive, encouraging automated AU detection and learning-based FER. In this work, we reference FACS/EMFACS primarily to motivate *landmark-aware* affect modeling: many AUs correspond to localized geometric changes (e.g., brow raising or lip corner pulling) that can be approximated using consistent landmark subsets and their relative configurations. This motivation becomes particularly relevant under occlusion, where texture/appearance cues are unreliable and geometry can provide a stable complementary signal. Representative AffectNet examples illustrating occlusion, pose variation, and illumination changes are shown in (Fig. 2).

The landmark correspondences in Table 1 are used as an illustrative geometric interpretation of AU-related facial regions. FACS itself defines muscle actions and does not prescribe indices from a particular 68-point landmark annotation scheme. GCA-ODN consequently does not use the listed correspondences as explicit AU supervision or as fixed anatomical constraints.

Automated AU detection has been explored using localized patches and multilabel learning to model AU co-occurrence (Zhao et al., 2015), region-specific classifiers (Jaiswal et al., 2015), and attention mechanisms under weak supervision (Shao et al., 2018a,b). While AU-based modeling can enhance interpretability (Tian et al., 2001), fixed AU taxonomies do not fully capture the diversity of expression behavior across individuals and contexts. This limitation becomes more pronounced under occlusion, where appearance cues may be missing and only partial geometric evidence remains.

Most modern FER systems rely on deep learning because learned representations capture subtle facial dynamics more effectively than hand-crafted features (Kennedy & Balint, 2016; Kowalski et al., 2017; Lopes et al., 2015). Convolutional neural networks (CNNs) dominate many



Fig. 2. Representative examples from AffectNet showing occlusion, pose variation, and illumination changes that can conceal or distort expression-discriminative facial evidence (Mollahosseini et al., 2017).

FER pipelines due to their ability to learn hierarchical spatial features and benefit from large-scale training data. Inception-style architectures (Szegedy et al., 2015), originally developed for large-scale image recognition (Deng et al., 2009a), have been adopted for FER via transfer learning and multi-scale feature extraction (Hasani & Mahoor, 2017).

Recent FER research increasingly combines local appearance learning with domain adaptation, geometric conditioning, self-supervised contrastive learning, and transformer-based relationship modeling (Liu et al., 2023a, 2024a,b, 2025; Peng et al., 2022; Xie et al., 2020). The Geometry-Aware Conditional Network captures structural facial patterns and long-range information to improve discrimination between visually confusable expressions (Liu et al., 2024b). Heatmap Neighbor Contrastive Learning constructs semantically related neighboring samples for self-supervised facial representation learning (Liu et al., 2025). MMATrans explicitly learns interactions among facial muscle-related regions through visual transformers, whereas FTATrans models texture-affinity relationships for video-based FER (Liu et al., 2026a, 2024a).

Occlusion-oriented FER methods can be broadly organized into occlusion recovery, facial-subregion analysis, and guidance from corresponding non-occluded images. Recovery-based methods attempt to reconstruct missing facial information, whereas subregion methods aggregate visible local regions and guidance-based methods transfer information from complete faces to occluded representations. A recent occlusion-aware low-rank framework integrates an explicit occlusion detector, masking, feature-level low-rank semantic recovery, and a paired non-occluded guidance branch (Wang et al., 2026). This framework differs from GCA-ODN in both scope and supervision: it is designed specifically for FER, requires explicit occlusion localization and paired clean-occluded guidance during training, and evaluates on RAF-DB, AffectNet, and dedicated synthetic or real-world occlusion benchmarks. GCA-ODN instead learns a shared FLD-FER representation without an independent occlusion-segmentation network or a paired clean-image branch. Consequently, the numerical results of the two frameworks are not directly transferable.

Compact CNN backbones such as ResNet-18 remain useful when local spatial precision, controlled capacity, and low-latency inference are design priorities (He et al., 2015). Transformer-based alternatives offer explicit modeling of distant facial-region relationships, but reported performance must be interpreted in light of differences in backbone capacity, pretraining data, input modality, dataset splits, and in-domain versus cross-dataset evaluation. For example, results obtained by training and testing on RAF-DB or AffectNet do not constitute the same protocol as training on AffectNet and directly evaluating CK+, JAFFE, or ISED without target-domain fine-tuning. This distinction is maintained in the FER comparisons in Section 6.

A related trend is observed in head-pose estimation, where orientation-aware and morphology-aware transformer models capture dependencies among facial regions during head rotation (Liu et al., 2023b, 2026b). Although these methods are not direct FLD or FER baselines, they provide additional evidence that modeling relationships among spatially separated facial regions can be useful under pose variation. Head-pose estimation is therefore discussed as a possible extension of the shared facial representation rather than as an experimentally evaluated task in this study.

2.3. Positioning of GCA-ODN

The reviewed literature addresses global context modeling, visibility reasoning, attention-based feature selection, geometric constraints, low-dimensional representations, and transformer-based relationship learning. However, these mechanisms are frequently developed for a single facial task, applied in separate processing stages, or evaluated under dataset-specific protocols. GCA-ODN coordinates global context aggregation, attention-guided dropout, and low-rank feature refinement within a shared embedding used by both FLD and FER heads. Its contribution lies in this coordinated joint-task formulation for partial observation, rather than in claiming that attention, dropout, global context, or low-rank learning is individually new.

The method is also distinguished from the two closely related 2026 frameworks discussed above. Relative to CAD-Net, GCA-ODN uses a global context module rather than recurrent criss-cross attention and

applies low-rank refinement to its shared representation rather than adopting CAD-Net's factorized landmark regression head (Sadiq et al., 2026b). Relative to the occlusion-aware FER framework, GCA-ODN does not require explicit occlusion segmentation, synthetic clean-occluded image pairs, or non-occluded feature-distribution guidance (Wang et al., 2026). These distinctions prevent architectural descriptions and experimental measurements from being conflated across the three models.

Motivated by these observations, this work emphasizes *landmark-aware* representation learning for joint FLD and FER under occlusion. Rather than relying solely on appearance-based classifiers or AU templates, we leverage geometric relationships and coordinated landmark movements as structured cues for affect inference. Building on this motivation, we propose a unified Global Context-Aware Dropout Network that combines global structural context, attention-guided suppression of corrupted evidence, and compact low-rank refinement to improve robustness under partial observation (as shown in Fig. 3, and Fig. 4).

3. Global context-aware dropout network (GCA-ODN)

To improve robustness under *partial observation* while maintaining practical efficiency, we propose the **Global Context-Aware Dropout Network (GCA-ODN)**. As shown in Fig. 3, GCA-ODN integrates three complementary mechanisms: (i) global context modeling to preserve long-range geometric dependencies, (ii) attention-guided dropout to suppress unreliable evidence during training, and (iii) low-rank refinement to obtain compact and noise-suppressed shared embeddings. A more detailed architectural view of the parallel Global Context and Attentive Dropout branches, followed by low-rank feature refinement and task heads, is provided in (Fig. 4). The design targets a common failure mode of occlusion-affected models: over-reliance on a small set of visible cues and unstable representations when informative regions are missing.

The architecture retains a compact convolutional backbone because dense landmark localization benefits from the local spatial inductive bias of CNNs. Long-range facial dependencies are introduced through the Global Context Module rather than by replacing the complete backbone with a transformer, thereby preserving a practical accuracy-latency balance for joint FLD and FER.

Let $Z \in \mathbb{R}^{B \times C \times H \times W}$ denote the feature map extracted from the backbone residual blocks, where B is the batch size and (C, H, W) are channel and spatial dimensions. GCA-ODN processes Z using two parallel branches:

$$Z^{gc} = GC(Z), \quad Z^{ad} = ADM(Z), \quad (1)$$

and then fuses the resulting features into a hybrid representation that is refined by a low-rank module:

$$\mathcal{X} = \phi([Z^{gc}; Z^{ad}]), \quad \mathcal{M} = LR(\mathcal{X}), \quad (2)$$

where $[\cdot; \cdot]$ denotes channel-wise concatenation, $\phi(\cdot)$ is a lightweight projection operator that aligns and fuses the concatenated feature channels, and $LR(\cdot)$ outputs a compact refined embedding $\mathcal{M}$ shared by the FLD and FER task heads. Spatial aggregation is performed inside the low-rank refinement stage rather than being included implicitly in both $\phi(\cdot)$ and $LR(\cdot)$. For clarity, Table 2 summarizes the main symbols used in this section.

In our formulation, occlusion is treated as *missing or corrupted evidence* rather than the absence of a facial attribute. Accordingly, GCA-ODN combines global structural cues through $GC(\cdot)$, attention-guided feature regularization through $ADM(\cdot)$, and compact refinement through $LR(\cdot)$ to reduce spurious dependence on occluded regions while preserving geometric coherence. The three components are optimized jointly; consequently, the proposed contribution lies in their coordinated use within a shared FLD-FER representation rather than in treating each constituent operation as independently novel.

GCA-ODN is architecturally distinct from CAD-Net and the separate occlusion-aware low-rank FER framework discussed in Section 2. It does

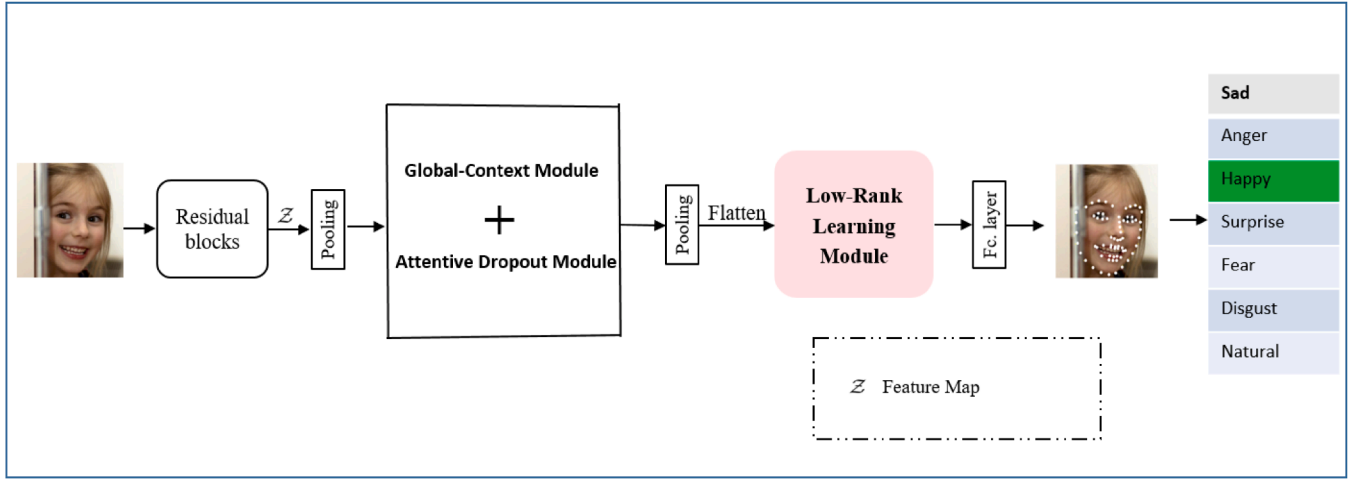


Fig. 3. Overview of GCA-ODN. The shared representation combines global context aggregation, attention-guided dropout, and low-rank feature refinement before being used for joint facial landmark detection and facial expression recognition.

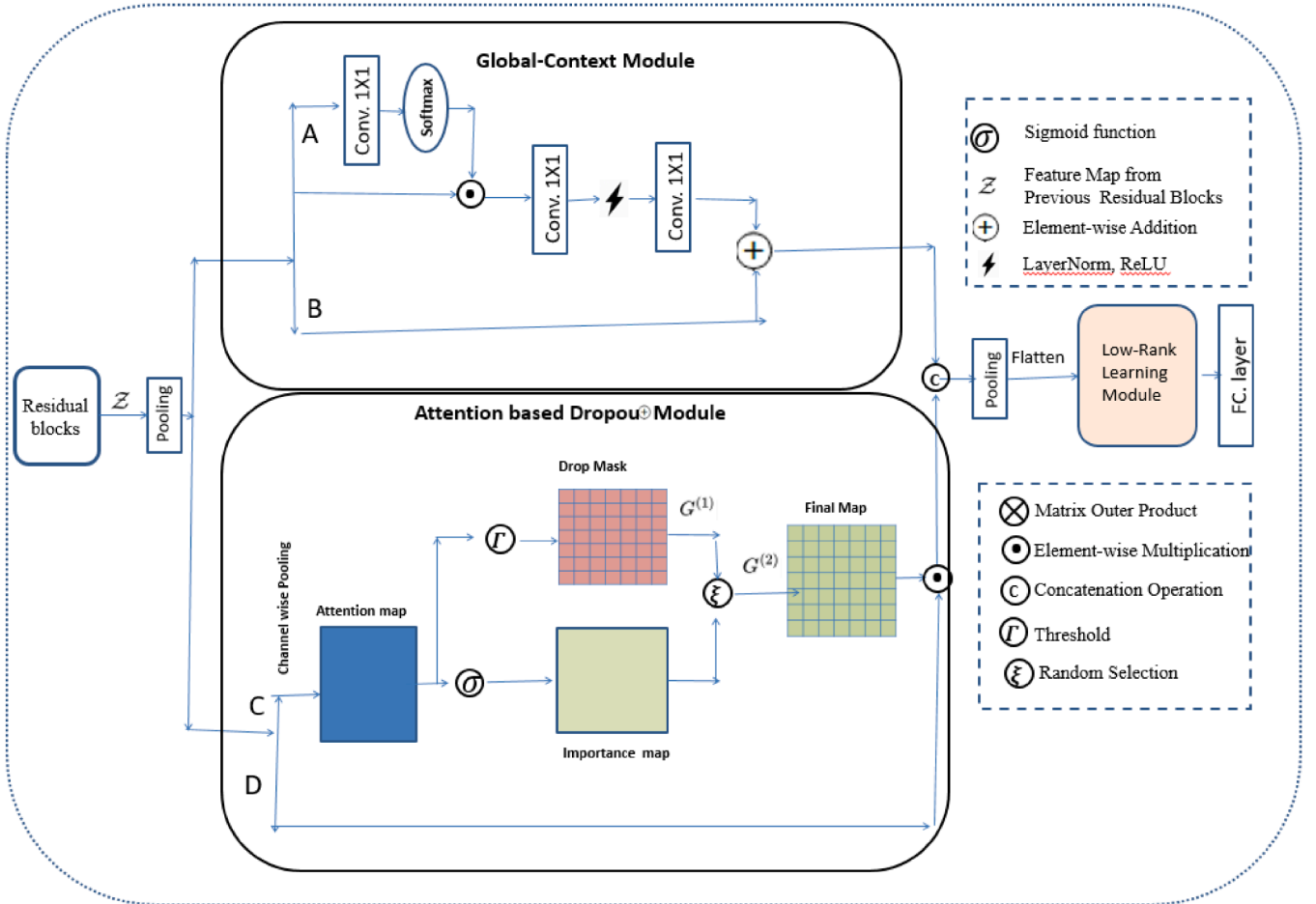


Fig. 4. Architecture of GCA-ODN. The shared CNN feature map is processed in parallel by the Global Context Module and Attentive Dropout Module. Their fused representation is subsequently compressed and denoised by the Low-Rank Feature Refinement Module before being supplied to the FLD and FER heads.

not employ recurrent criss-cross attention, a factorized landmark-only regression head, an external pixel-level occlusion detector, or a paired clean-occluded guidance branch. Therefore, the module definitions and experimental measurements of these related frameworks are not used interchangeably with those of GCA-ODN (Sadiq et al., 2026b; Wang et al., 2026).

3.1. Global context module

The Global Context Module is inspired by GCNet (Cao et al., 2019) and aggregates long-range spatial information into a compact descriptor that is redistributed through a lightweight channel transformation. Given $Z \in \mathbb{R}^{B \times C \times H \times W}$, a 1×1 convolution produces a spatial score map

Table 2

Notation used to describe the GCA-ODN architecture and its three constituent modules.

Symbol	Description
Z	Backbone feature map of size $B \times C \times H \times W$
Z^{gc}	Feature map produced by the Global Context branch
Z^{ad}	Feature map produced by the Attentive Dropout branch
$\mathcal{X}$	Fused representation after projection $\phi(\cdot)$
$\bar{\mathcal{X}}$	Spatially aggregated fused representation
$\mathcal{M}$	Shared embedding produced by low-rank refinement
ω_i	Normalized global-context weight at spatial position i
U	Channel-averaged ADM attention response
$\tilde{U}$	Min-max-normalized ADM response
D	Binary ADM drop mask
P	Soft ADM importance map
τ_{adm}	Threshold used to construct the ADM drop mask
p	Probability of selecting the ADM drop-mask mode
r_{gc}	Channel-reduction ratio in the Global Context transform
r_A	Maximum rank of the low-dimensional LR mapping
γ	Learnable scale for residual context injection
A	Learnable low-rank mapping used to obtain $\mathcal{M}$
Q_A, R_A	Low-dimensional factors used to parameterize A

$S \in \mathbb{R}^{B \times 1 \times H \times W}$, which is normalized with softmax to obtain attention weights:

$$\omega_i = \frac{\exp(S_i)}{\sum_{j=1}^{H \times W} \exp(S_j)}, \quad \sum_{i=1}^{H \times W} \omega_i = 1, \quad (3)$$

where i indexes spatial locations. A global context vector is computed by attention-weighted pooling:

$$g = \sum_{i=1}^{H \times W} \omega_i Z_i, \quad g \in \mathbb{R}^{B \times C \times 1 \times 1}. \quad (4)$$

The context vector is then transformed using two 1×1 convolutions with a nonlinear activation:

$$t = W_2 \delta(W_1 g), \quad (5)$$

where

$$W_1 : \mathbb{R}^C \rightarrow \mathbb{R}^{C/r_{gc}}, \quad W_2 : \mathbb{R}^{C/r_{gc}} \rightarrow \mathbb{R}^C. \quad (6)$$

The transformed context is fused back into the original feature map through residual addition:

$$Z^{gc} = Z + \gamma t, \quad (7)$$

where $\delta(\cdot)$ denotes ReLU and γ is a learnable scaling parameter. This residual context injection allows each spatial position to be informed by the global facial structure, improving geometric consistency under occlusion, pose, and illumination changes.

Unlike the recurrent criss-cross operation used in CAD-Net, which propagates information along horizontal and vertical axes, the present module obtains a single attention-weighted global descriptor and redistributes the transformed context to all spatial locations (Sadiq et al., 2026b). The two context mechanisms should therefore not be described as the same block or assigned the same ablation and efficiency measurements.

The bottleneck transformation limits the additional channel-processing cost. The channel-reduction ratio is fixed at $r_{gc} = 8$ in all reported GCA-ODN experiments. Candidate ratios of 8, 16, 32, and 64 were considered during configuration selection, but no unmeasured sensitivity values are inferred or transferred from CAD-Net.

3.2. Attentive dropout module

The Attentive Dropout Module (ADM) extends standard dropout by using an attention-derived mask to reduce repeated dependence on dominant spatial responses during training. Building on attention-guided dropout (Choe et al., 2020; Choe & Shim, 2019; Sadiq & Shi, 2022; Sadiq

et al., 2019), ADM generates two complementary spatial maps from an attention summary: a *drop mask* that removes dominant responses to prevent over-reliance on a small set of salient cues, and an *importance map* that softly emphasizes informative evidence.

Given Z , we compute a spatial attention response through channel-wise average pooling:

$$U = \text{AvgPool}_c(Z) \in \mathbb{R}^{B \times 1 \times H \times W}. \quad (8)$$

To make the binary threshold independent of the absolute activation scale, the response is normalized separately for each sample:

$$\tilde{U} = \frac{U - \min_{h,w}(U)}{\max_{h,w}(U) - \min_{h,w}(U) + \epsilon}, \quad (9)$$

where ϵ is a small positive constant used for numerical stability.

A binary drop mask is produced by thresholding the normalized response:

$$D = \mathbb{1}[\tilde{U} < \tau_{adm}], \quad \tau_{adm} = 0.5. \quad (10)$$

The threshold is fixed across all datasets and experiments. Using the distinct symbol τ_{adm} also prevents confusion with the low-rank regularization coefficients introduced in Section 4.

In parallel, the soft importance map is computed from the unthresholded attention response:

$$P = \sigma(U), \quad (11)$$

where $\sigma(\cdot)$ denotes the sigmoid function. Unlike the binary drop mask, the importance map retains continuously valued spatial weights and therefore preserves graded evidence from visible facial regions.

During training, ADM alternates between the two modes:

$$Z^{ad} = \begin{cases} Z \odot D, & \text{with probability } p, \\ Z \odot P, & \text{with probability } 1 - p, \end{cases} \quad p = 0.5, \quad (12)$$

where $\odot$ denotes element-wise multiplication. The drop-mask mode prevents the network from collapsing onto a small set of dominant cues that may be absent under occlusion, while the importance-map mode reinforces reliable evidence. The balanced mode probability assigns equal training frequency to the drop-mask and importance-map operations. The two modes perform different functions: the first diversifies the spatial evidence used during learning, whereas the second retains soft spatial weighting rather than applying indiscriminate random dropout.

During inference, stochastic mode selection and binary feature dropping are disabled. The deterministic importance-map operation is used instead:

$$Z_{\text{test}}^{ad} = Z \odot P. \quad (13)$$

This deterministic formulation avoids randomly suppressing facial evidence at test time and ensures that repeated inference on the same input produces the same ADM output.

ADM is an internal feature-space regularizer rather than an explicit occlusion detector. It neither produces pixel-level occlusion annotations nor requires corresponding clean and synthetically occluded image pairs, which distinguishes it from detector–masking–recovery FER pipelines (Wang et al., 2026). The settings $\tau_{adm} = 0.5$ and $p = 0.5$ are specific to GCA-ODN and are held fixed across the reported datasets; they are not copied from CAD-Net, ADODN, or another attentive-dropout implementation.

3.3. Low-rank feature refinement module

The Low-Rank Feature Refinement Module enforces compactness and suppresses redundancy and noise in the fused representation $\mathcal{X}$ from Eq. (2). Under occlusion, correlations among visible facial components can be exploited to produce a more stable shared representation. The fused tensor is first spatially aggregated and reshaped as

$$\bar{\mathcal{X}} = \text{Pool}(\mathcal{X}) \in \mathbb{R}^{B \times d}. \quad (14)$$

Table 3
Fixed module settings used in the reported GCA-ODN experiments.

Setting	Value or operation
GC reduction ratio	$r_{gc} = 8$
ADM threshold	$\tau_{adm} = 0.5$
ADM drop-mask probability	$p = 0.5$
ADM importance-map probability	$1 - p = 0.5$
Low-rank dimension	$r_A = 8$
Training-time ADM	Stochastic selection between D and P
Inference-time ADM	Deterministic importance weighting, $Z \odot P$

To avoid a notation collision between the ADM response U and the factors of the low-rank mapping, the refinement matrix is parameterized using Q_A and R_A :

$$A = Q_A R_A^T, \quad Q_A, R_A \in \mathbb{R}^{d \times r_A}, \quad r_A \ll d. \quad (15)$$

The shared representation is then obtained as

$$\mathcal{M} = \bar{\mathcal{X}} A, \quad \mathcal{M} \in \mathbb{R}^{B \times d}. \quad (16)$$

The factorization guarantees

$$\text{rank}(A) \leq r_A, \quad (17)$$

and permits an SVD-free variational representation of the nuclear-norm regularizer:

$$\|A\|_* = \min_{A=Q_A R_A^T} \frac{1}{2} (\|Q_A\|_F^2 + \|R_A\|_F^2). \quad (18)$$

Accordingly, the effective rank is controlled through r_A , while the Frobenius penalties on Q_A and R_A provide a differentiable surrogate for nuclear-norm regularization. This formulation avoids performing singular value decomposition at every optimization step and is consistent with the objective in Section 4 (Zhu et al., 2019).

The rank is fixed at $r_A = 8$ in all reported experiments. This choice applies the strongest compactness constraint among the considered candidate ranks $r_A \in \{8, 16, 32, 64\}$ and keeps the refinement mapping lightweight. No numerical rank-sensitivity results are claimed unless they are supported by the corresponding GCA-ODN measurements.

The present module refines the fused representation before it is supplied to both task heads. This differs from CAD-Net, where low-rank factorization is applied to the landmark-regression mapping itself; consequently, the rank values, complexity measurements, and sensitivity results of the two networks are architecture-specific (Sadiq et al., 2026b).

3.4. Interplay of the three modules

The three modules are designed to be complementary. The Global Context branch preserves holistic geometric structure and long-range dependencies; the Attentive Dropout branch reduces dependence on dominant or potentially corrupted responses during learning; and the Low-Rank refinement compresses and denoises the fused representation to stabilize the shared embedding.

In particular, GCM supplies contextual information that can compensate for locally missing evidence, ADM encourages the network to distribute attention beyond its strongest visible regions, and LM restricts the fused representation to a compact subspace. Their functions are therefore not directly substitutable. The available component study evaluates this complementarity through controlled leave-one-module-out configurations in Section 7. It does not, by itself, constitute an isolated evaluation of every internal drop-mask, importance-map, or low-rank sub-component.

Unless otherwise stated, all GCA-ODN experiments use $r_{gc} = 8$, $\tau_{adm} = 0.5$, $p = 0.5$, and $r_A = 8$. Stochastic ADM mode selection is used only during training, whereas deterministic importance-map weighting is used during inference. These settings remain unchanged across datasets and are specific to GCA-ODN. Together, these mechanisms improve robustness under partial observation for joint FLD and FER.

4. Optimization of proposed methodology

This section describes the end-to-end optimization of **GCA-ODN** for *joint* facial landmark detection (FLD) and facial expression recognition (FER) under occlusion. Given a training set $D = \{(I_n, \hat{S}_n, \hat{c}_n)\}_{n=1}^M$, I_n denotes an input face image, $\hat{S}_n \in \mathbb{R}^{L \times 2}$ denotes the ground-truth landmark coordinates, and $\hat{c}_n \in \{1, \dots, C_e\}$ is the ground-truth emotion label. Either annotation may be unavailable for a particular sample. The network, parameterized by Θ , predicts landmarks $\hat{S}_n$ and emotion logits $\hat{p}_n$ through two task heads operating on the shared embedding $\mathcal{M}$ (Section 3).

To support datasets containing only landmark annotations, only expression annotations, or both, we introduce binary label-availability indicators a_n^{FLD} and a_n^{FER} . An indicator equals one when the corresponding annotation is available and zero otherwise. For a training batch, we define

$$M_{\text{FLD}} = \sum_{n=1}^M a_n^{\text{FLD}}, \quad M_{\text{FER}} = \sum_{n=1}^M a_n^{\text{FER}}, \quad (19)$$

and use the protected denominators

$$\bar{M}_{\text{FLD}} = \max(1, M_{\text{FLD}}), \quad \bar{M}_{\text{FER}} = \max(1, M_{\text{FER}}). \quad (20)$$

When a mini-batch contains no valid annotation for one task, the corresponding indicator-weighted sum is zero and that task term vanishes without producing a division-by-zero error. This formulation supports joint, FLD-only, and FER-only optimization without assigning artificial labels to incompletely annotated samples.

Let A denote the learnable mapping associated with the Low-Rank Feature Refinement Module. Consistent with Section 3, we parameterize this mapping as

$$A = Q_A R_A^T, \quad Q_A, R_A \in \mathbb{R}^{d \times r_A}, \quad r_A = 8. \quad (21)$$

The factors Q_A and R_A are included in Θ . We further denote by $G_n^{(1)}$ and $G_n^{(2)}$ the differentiable ADM responses associated with the drop-mask and importance-map modes, respectively. These responses are regularized instead of directly applying a gradient-based penalty to the non-differentiable binary mask.

The unified objective combines the task losses with regularization terms that promote compactness, numerical stability, weight control, and spatial selectivity:

$$\begin{aligned} \min_{\Theta} \mathcal{L}_{\text{GCA-ODN}} = & \frac{\lambda_{\text{FLD}}}{M_{\text{FLD}}} \sum_{n=1}^M a_n^{\text{FLD}} \mathcal{L}_{\text{FLD}}(\hat{S}_n, \hat{S}_n) \\ & + \frac{\lambda_{\text{FER}}}{M_{\text{FER}}} \sum_{n=1}^M a_n^{\text{FER}} \mathcal{L}_{\text{FER}}(\hat{p}_n, \hat{c}_n) \\ & + \frac{\lambda_*}{2} (\|Q_A\|_F^2 + \|R_A\|_F^2) + \rho \|Q_A R_A^T\|_F^2 \\ & + \alpha_c \|K_c\|_F^2 + \alpha_f \|K_f\|_F^2 \\ & + \frac{\kappa_1}{M} \sum_{n=1}^M \|G_n^{(1)}\|_1 + \frac{\kappa_2}{M} \sum_{n=1}^M \|G_n^{(2)}\|_1, \end{aligned} \quad (22)$$

where K_c and K_f denote the convolutional and fully connected weights, respectively. The factor penalty weighted by λ_* is the SVD-free variational surrogate for the nuclear norm, whereas ρ regularizes the reconstructed mapping $A = Q_A R_A^T$. The ADM terms and task losses are averaged over their applicable samples so that their scale does not grow with the number of training examples.

For a jointly annotated batch, both task weights in Table 4 are active. For an FLD-only or FER-only batch, the unavailable task term becomes zero through its label-availability indicators; the remaining task weight is not renormalized. The regularization of K_c and K_f is implemented through the reported optimizer weight decay and is not added a second time as an independent software operation.

Table 4

Task-loss weights and regularization coefficients used in the reported GCA-ODN configuration. Task losses are activated only for samples with the corresponding annotations.

Term	Value	Description
λ_{FLD}	0.4	FLD loss weight during joint optimization
λ_{FER}	0.6	FER loss weight during joint optimization
λ_*	1×10^{-5}	factorized nuclear-norm surrogate weight
ρ	1×10^{-5}	Frobenius penalty on the reconstructed mapping A
α_c	5×10^{-4}	regularization coefficient for convolutional weights
α_f	5×10^{-4}	regularization coefficient for fully connected weights
κ_1	1×10^{-5}	sparsity coefficient for the drop-mode response
κ_2	1×10^{-5}	sparsity coefficient for the importance-map response

The fixed architecture settings used throughout optimization are $r_{\text{gc}} = 8$, $\tau_{\text{adm}} = 0.5$, $p = 0.5$, and $r_A = 8$, as summarized in Table 3. These values are held constant across the reported datasets.

4.1. Task losses

For FLD, we use the average squared localization error over the annotated landmarks:

$$\mathcal{L}_{\text{FLD}}(\hat{S}_n, \check{S}_n) = \frac{1}{L} \|\hat{S}_n - \check{S}_n\|_F^2. \quad (23)$$

The factor $1/L$ prevents the loss scale from changing when datasets use different numbers of annotated landmarks. The coordinates are expressed in the normalized input coordinate system during training, whereas benchmark evaluation uses the dataset-specific normalization defined in Section 5.2.

For FER, we use cross-entropy over the emotion classes:

$$\mathcal{L}_{\text{FER}}(\hat{p}_n, \check{c}_n) = -\log(\text{softmax}(\hat{p}_n)_{\check{c}_n}). \quad (24)$$

The softmax operation is applied to the FER logits only for the cross-entropy calculation; the landmark-regression head does not share this output transformation.

4.2. Low-rank regularization without explicit SVD

Directly minimizing $\text{Rank}(A)$ is non-convex. We therefore use the nuclear norm as a convex surrogate and adopt its variational factorization:

$$\|A\|_* = \min_{A=Q_A R_A^T} \frac{1}{2} (\|Q_A\|_F^2 + \|R_A\|_F^2). \quad (25)$$

Here,

$$Q_A, R_A \in \mathbb{R}^{d \times r_A}, \quad r_A = 8, \quad \text{rank}(A) \leq r_A. \quad (26)$$

Eq. (22) directly optimizes the factorized representation, so an explicit singular value decomposition of A is not required at each training iteration. The candidate ranks considered during configuration selection were $r_A \in \{8, 16, 32, 64\}$, and the reported model uses $r_A = 8$. No numerical rank-sensitivity values are inferred from CAD-Net or another low-rank architecture.

The factorized mapping is applied to the fused shared representation before the FLD and FER task heads. It therefore differs from low-rank formulations applied only to a landmark-regression matrix or only to an FER recovery branch.

4.3. Sparsity of attention-guided responses

The binary drop mask D is obtained through thresholding and is not itself used as the target of a gradient-based sparsity penalty. Instead, we regularize differentiable responses associated with the two ADM modes. For the drop-mask pathway, the continuous response above the threshold is defined as

$$G_n^{(1)} = \text{ReLU}(\tilde{U}_n - \tau_{\text{adm}}), \quad \tau_{\text{adm}} = 0.5. \quad (27)$$

Algorithm 1 End-to-end optimization of GCA-ODN with label-aware task losses, balanced ADM mode selection, and SVD-free low-rank refinement.

Require: Training set $\mathcal{D}$, learning rate η , $\tau_{\text{adm}} = 0.5$, $p = 0.5$, $r_A = 8$, and the loss weights in Table 4

```

1: Initialize  $\Theta$ , including  $Q_A$  and  $R_A$ 
2: for each mini-batch  $\mathcal{B}$  do
3:   Read indicators  $a_n^{\text{FLD}}$  and  $a_n^{\text{FER}}$ 
4:   Compute  $\bar{M}_{\text{FLD}}$  and  $\bar{M}_{\text{FER}}$ 
5:    $Z \leftarrow f_{\text{backbone}}(I_n)$ 
6:    $Z^{\text{gc}} \leftarrow \text{GC}(Z)$ 
7:    $U \leftarrow \text{AvgPool}_c(Z)$ 
8:   Normalize  $U$  to obtain  $\tilde{U}$ 
9:    $D \leftarrow \mathbb{1}[\tilde{U} < \tau_{\text{adm}}]$ 
10:   $P \leftarrow \sigma(U)$ 
11:  Sample  $b \sim \text{Bernoulli}(p)$ 
12:  if  $b = 1$  then
13:     $Z^{\text{ad}} \leftarrow Z \odot D$ 
14:  else
15:     $Z^{\text{ad}} \leftarrow Z \odot P$ 
16:  end if
17:   $\mathcal{X} \leftarrow \phi([Z^{\text{gc}}; Z^{\text{ad}}])$ 
18:   $\bar{\mathcal{X}} \leftarrow \text{Pool}(\mathcal{X})$ 
19:   $A \leftarrow Q_A R_A^T$ 
20:   $\mathcal{M} \leftarrow \bar{\mathcal{X}} A$ 
21:   $\hat{S}_n \leftarrow h_{\text{FLD}}(\mathcal{M})$ 
22:   $\hat{p}_n \leftarrow h_{\text{FER}}(\mathcal{M})$ 
23:  Compute only task losses with available annotations
24:   $G_n^{(1)} \leftarrow \text{ReLU}(\tilde{U}_n - \tau_{\text{adm}})$ 
25:   $G_n^{(2)} \leftarrow P_n$ 
26:  Compute  $\mathcal{L}_{\text{GCA-ODN}}$  using Eq. (22)
27:   $\Theta \leftarrow \text{AdamUpdate}(\Theta, \nabla_{\Theta} \mathcal{L}_{\text{GCA-ODN}}, \eta)$ 
28: end for

```

For the importance-map pathway, the regularized response is

$$G_n^{(2)} = P_n = \sigma(U_n). \quad (28)$$

To promote sparse and spatially selective responses, we apply an L_1 penalty:

$$\|G\|_1 = \sum_k |g_k|, \quad G \in \{G_n^{(1)}, G_n^{(2)}\}. \quad (29)$$

The first term penalizes the magnitude and spatial extent of responses that exceed the drop threshold, whereas the second discourages a uniformly active importance map. The two responses are regularized separately because the drop-mode response and soft importance map have different activation characteristics. Their coefficients are kept equal in the reported configuration so that neither mode is favored through unequal regularization.

4.4. End-to-end training

All parameters are optimized jointly with mini-batch Adam using the learning-rate schedule reported in Table 6. During training, ADM alternates between the drop-mask and importance-map modes with probability $p = 0.5$ (Section 3.2), balancing suppression of dominant cues with soft reinforcement of spatial evidence. A single ADM mode is selected for each forward pass, and the same resulting shared embedding is supplied to both task heads. The FLD and FER predictions are therefore optimized from a consistent feature realization (Algorithm 1).

At inference, stochastic ADM mode selection and binary feature dropping are disabled. The deterministic importance-map operation is used:

$$Z_{\text{test}}^{\text{ad}} = Z \odot P. \quad (30)$$

Table 5

Datasets and evaluation roles used in the FLD experiments. COFW is evaluated using its 68-point re-annotated test set, whereas AFLW follows the standard 19-landmark protocol.

Dataset	Training split	Test split	#Landmarks	Primary role	Main challenge
300W (Sagonas et al., 2013)	3,148 images	689 images	68	train/test	occlusion and pose
COFW (Burgos-Artizzu et al., 2013; Ghiasi & Fowlkes, 2014)	–	507 images	68	cross-dataset test	severe occlusion
AFLW (Koestinger et al., 2011)	20,000 faces	4,386 faces	19	test	large pose variation
300VW (Tzimiropoulos, 2015)	50 videos	64 videos	68	train/test	temporal variation
Menpo (Zafeiriou et al., 2017)	7,564 images	–	68	additional pretraining	appearance diversity

The resulting embedding is then processed by the FLD and FER heads without stochastic feature suppression. This produces deterministic predictions for a fixed input and model state.

The factorized nuclear-norm term promotes a compact latent subspace in the refined representation, the Frobenius and weight penalties stabilize learning, and the L_1 terms encourage spatially selective ADM responses. Together with the task losses in Eqs. (23) and (24), these components enable stable end-to-end optimization of GCA-ODN for joint FLD and FER under occlusion. The formulation removes sample-index ambiguity from the ADM regularization terms, avoids penalizing the non-differentiable binary mask directly, makes the treatment of partially annotated datasets explicit, and remains consistent with the fixed GCA-ODN settings defined in Section 3.

5. Experimental details for facial landmark detection

This section evaluates GCA-ODN on widely used facial landmark detection (FLD) benchmarks under normal conditions, severe occlusion, large pose variations, and video settings. We first describe datasets and evaluation protocols (Section 5.1), then report metrics and implementation details to support reproducibility (Section 5.2). Finally, we present results under normal conditions, occlusion, pose variations, and video sequences (Sections 5.3–5.3.4). Unless otherwise stated, all faces are cropped and resized to 224×224 , and the backbone is initialized with ImageNet-pretrained weights (Deng et al., 2009b).

5.1. Datasets and specifications

We conduct experiments on five standard FLD datasets: 300W (Sagonas et al., 2013), COFW (Burgos-Artizzu et al., 2013), AFLW (Koestinger et al., 2011), Menpo (Zafeiriou et al., 2017), and 300VW (Tzimiropoulos, 2015). The training and testing roles, landmark protocols, and main evaluation challenges of these datasets are summarized in (Table 5). These benchmarks cover diverse factors, including severe occlusion, large pose variation, illumination changes, and temporal dynamics. We compare against representative regression-, heatmap-, visibility-, and transformer-based methods (Browatzki & Wallraven, 2020; Gao et al., 2020; Sadiq & Shi, 2022; Wan et al., 2023; Xiang et al., 2025; Zhu et al., 2019). The comparison additionally includes ADODN, CAD-Net, and Proto-Former as recent methods. Published results obtained using different architectures, additional facial data, or multi-dataset training are explicitly marked and are not interpreted as strictly controlled comparisons.

Training regimes. For 300W, the base GCA-ODN model is trained using the official 300W training split. For AFLW and 300VW, the corresponding benchmark-specific training splits are used. COFW is treated as a cross-dataset evaluation using the 68-point re-annotated test set, without COFW fine-tuning. GCA-ODN(M) uses the same architecture and target-benchmark protocol as GCA-ODN but incorporates additional Menpo pretraining. The two variants are therefore reported separately so that improvements due to additional facial training data are not conflated with the architecture itself.

300W (Sagonas et al., 2013) contains 3837 images collected from AFW, LFPW, and HELEN. We use 3148 images for training and 689 im-

ages for testing, with the standard division into *Common* (554 images), *Challenging* (135 IBUG images), and *Full* (all 689 images).

COFW (Burgos-Artizzu et al., 2013) contains 1852 images with substantial occlusions and diverse expressions and poses. The original dataset provides 29 landmarks. For comparison with 68-point methods, we use the publicly available 68-point re-annotation of its 507-image test set (Ghiasi & Fowlkes, 2014). No COFW images are used to fine-tune GCA-ODN in this evaluation.

AFLW (Koestinger et al., 2011) includes 21,997 images containing 25,993 annotated faces collected from Flickr. Following the commonly used evaluation protocol, the two ear landmarks are excluded, resulting in 19 evaluated landmarks. We use 20,000 training faces, the 4,386-face AFLW-Full test set, and the 1,165-face frontal subset for AFLW-Frontal evaluation.

300VW (Tzimiropoulos, 2015) contains 114 videos with per-frame 68-landmark annotations. We use the official 50-video training split and 64 test videos, which are grouped into Categories 1–3 according to increasing difficulty.

Menpo (Zafeiriou et al., 2017) provides near-frontal and profile facial annotations. Since the official test annotations are not publicly available, the 7564 near-frontal training images with 68-point annotations are used only as additional facial pretraining data. The corresponding variant is denoted as **GCA-ODN(M)**.

5.2. Metrics for evaluation and technical implementation details

Metrics. We report the mean normalized Euclidean landmark error, denoted as NRMSE to remain consistent with the terminology used in the ODN-family literature, together with Cumulative Error Distribution curves. Let $\tilde{s}_{i,k} \in \mathbb{R}^2$ and $s_{i,k} \in \mathbb{R}^2$ denote the ground-truth and predicted coordinates of the k th landmark for the i th test image. The reported error is computed as

$$\text{NRMSE} = \frac{1}{N} \sum_{i=1}^N \frac{1}{L} \sum_{k=1}^L \frac{\|\tilde{s}_{i,k} - s_{i,k}\|_2}{\Omega_i}, \quad (31)$$

where N is the number of test images, L is the number of landmarks, and Ω_i is a face-scale normalization term. For 300W, COFW, and 300VW, Ω_i is the inter-ocular distance; for AFLW, face bounding-box width is used. CED reports the proportion of test samples whose normalized error falls below each threshold. Although the historical term NRMSE is retained in the tables, Eq. (31) is a mean normalized Euclidean error and does not apply an additional square or square-root operation.

Preprocessing and augmentation. Faces are cropped using the provided bounding boxes, or detections where required, and resized to 224×224 . Training augmentation includes random rotation ($\pm 30^\circ$), scaling (0.7 – $1.3\times$), translation of up to 10%, and horizontal flipping with probability 0.5. Inputs are normalized using ImageNet channel statistics. The backbone is initialized from ImageNet-pretrained weights (Deng et al., 2009b).

Optimization settings. We use Adam with an initial learning rate of 1×10^{-4} and batch size 32. The learning rate is multiplied by 0.1 every 30 epochs, and training runs for 120 epochs. Weight decay is set to 5×10^{-4} . All experiments are implemented in PyTorch and executed on an NVIDIA RTX A6000 GPU.

Table 6
Verified training, module, and runtime settings used in the FLD experiments.

Item	Setting
Input size	224×224
Backbone	ResNet-18
Backbone initialization	ImageNet-pretrained weights (Deng et al., 2009b)
Optimizer	Adam
Initial learning rate	1×10^{-4}
Batch size	32
Training epochs	120
Learning-rate schedule	multiply by 0.1 every 30 epochs
Weight decay	5×10^{-4}
Augmentation	rotation $\pm 30^\circ$, scale 0.7–1.3 $\times$, translation $\leq 10\%$, flip probability 0.5
ADM response normalization	per-map min–max normalization to [0, 1]
ADM threshold	$\tau_{\text{adm}} = 0.5$
ADM mode probability	$p = 0.5$
Inference-time ADM	deterministic importance-map weighting
GC reduction ratio	$r_{\text{gc}} = 8$
Low-rank factorization rank	$r_A = 8$
Hardware	NVIDIA RTX A6000
Inference setting	input 224×224 , batch size 1
Verified inference speed	> 80 FPS

Table 7

Consolidated GCA-ODN results across the evaluated FLD benchmarks. Values are NRMSE ($\times 10^{-2}$) and are reproduced from Tables 8–11.

Model	300W-C	300W-F	300W-Ch	AFLW-Full	AFLW-Frontal	300VW-C1	300VW-C2	300VW-C3
GCA-ODN	2.60	3.20	5.61	1.30	1.03	4.40	4.01	6.28
GCA-ODN(M)	2.56	3.14	5.56	1.26	1.01	4.31	4.01	6.21

Module hyperparameters. Unless otherwise stated, the regularization weights in Table 4 are used. Before ADM thresholding, the spatial response is min–max normalized to [0, 1]. The fixed GCA-ODN configuration uses $\tau_{\text{adm}} = 0.5$, $p = 0.5$, $r_{\text{gc}} = 8$, and $r_A = 8$. Stochastic switching between the ADM drop-mask and importance-map modes is applied only during training. During inference, the deterministic importance-map operation is used, as defined in Eq. (13).

These hyperparameters are specific to GCA-ODN. Values and sensitivity measurements reported for CAD-Net, ADODN, or other attentive-dropout and low-rank architectures are not transferred to this model. Numerical sensitivity claims are restricted to GCA-ODN measurements that are explicitly reported in the ablation section.

Efficiency and reproducibility. Table 6 summarizes the verified implementation settings. Inference speed is measured at an input resolution of 224×224 with batch size 1 on an NVIDIA RTX A6000. Under this setting, GCA-ODN processes more than 80 frames per second. This measurement establishes real-time throughput only for the stated hardware and input configuration. Parameter count, FLOPs, peak memory, and latency values must be obtained from the final GCA-ODN implementation and are not inferred from the published measurements of CAD-Net or another related architecture.

5.3. Empirical analysis

5.3.1. Evaluation under normal conditions

We first evaluate under relatively mild conditions using the 300W Common and Full sets. Table 8 reports NRMSE ($\times 10^{-2}$). GCA-ODN achieves 2.60 on the Common set and 3.20 on the Full set. Relative to RHT-R, the Common-set error is reduced by approximately 9.4%, whereas the Full-set error is approximately 5.3% lower than the reported POPos result. Additional Menpo pretraining reduces the errors to 2.56 and 3.14, respectively.

Fig. 5 shows the corresponding CED curves. The curve-level comparison complements the average error and indicates how localization performance is distributed across the test images.

Table 8

Comparison on the 300W Common and Full sets (NRMSE $\times 10^{-2}$).

Method	Year	Common	Full
ODN (Zhu et al., 2019)	2019	3.56	4.17
ADN (Sadiq et al., 2019)	2019	3.52	4.14
LGSA (Gao et al., 2020)	2020	3.36	4.06
3FabRec (Browatzki & Wallraven, 2020)	2020	3.36	3.82
AODN (Sadiq & Shi, 2022)	2022	3.27	3.76
RHT-R (Wan et al., 2023)	2023	2.87	3.46
POPpos (Xiang et al., 2025)	2025	–	3.38
ADODN (Sadiq et al., 2026a)	2026	2.80	3.35
GCA-ODN	2026	2.60	3.20
GCA-ODN(M)	2026	2.56	3.14

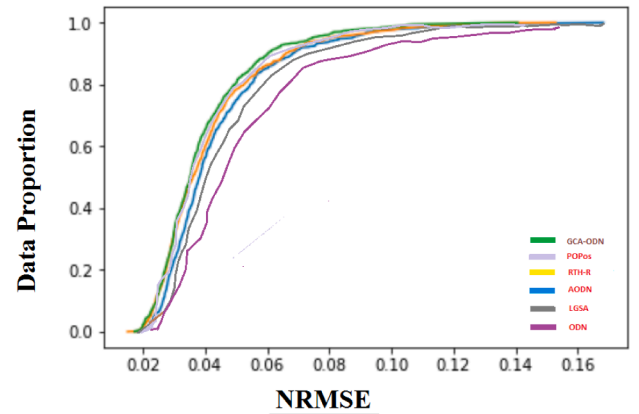


Fig. 5. Cumulative Error Distribution curves on the 300W test set. Curves closer to the upper-left indicate that a larger proportion of images is localized below a given normalized-error threshold.

5.3.2. Evaluation under occlusion

To assess robustness under heavy occlusion, we evaluate on the 300W Challenging subset and COFW. Table 9 shows that GCA-ODN reduces the error relative to the listed ODN-family and occlusion-aware

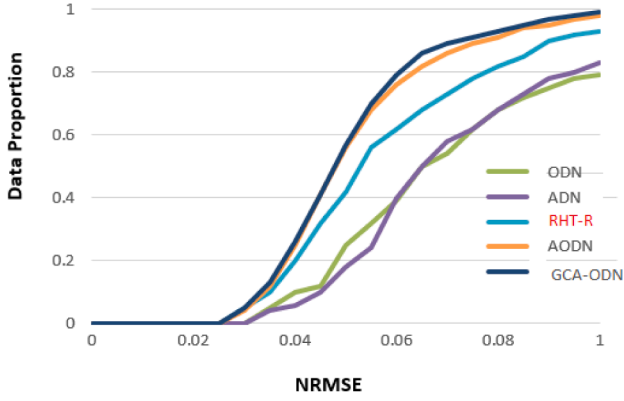


Fig. 6. Cumulative Error Distribution curves on the 300W Challenging subset under severe occlusion and pose variation.

Table 9

Comparison on the 300W Challenging subset (NRMSE $\times 10^{-2}$). † Proto-Former uses a multi-dataset transformer protocol, while CAD-Net uses a distinct geometry-aware architecture and training configuration. These rows provide contemporary context rather than strictly identical-data comparisons.

Method	Year	Challenging
ODN (Zhu et al., 2019)	2019	6.67
ADN (Sadiq et al., 2019)	2019	6.60
RetinaFace (Deng et al., 2020)	2020	6.83
AODN (Sadiq & Shi, 2022)	2022	6.38
RHT-R (Wan et al., 2023)	2023	5.88
ADODN (Sadiq et al., 2026a)	2026	5.81
CAD-Net† (Sadiq et al., 2026b)	2026	5.21
Proto-Former† (Hu et al., 2026)	2026	4.39
GCA-ODN	2026	5.61
GCA-ODN(M)	2026	5.56

CNN baselines. Compared with RHT-R and ADODN, GCA-ODN reduces NRMSE by approximately 4.6% and 3.4%, respectively.

CAD-Net and Proto-Former report lower errors under their respective experimental protocols. CAD-Net uses recurrent criss-cross geometry modeling and a factorized landmark-regression head, whereas Proto-Former adopts prototype-aware multi-dataset transformer training. GCA-ODN is therefore positioned as a joint FLD-FER model with real-time throughput under its stated hardware setting rather than as the lowest-error method across every architecture and training regime.

Fig. 6 presents the CED curves on 300W Challenging. These curves are interpreted together with the mean normalized error and do not substitute for repeated-run or paired statistical measurements.

COFW contains irregular occlusions over the eyes, mouth, facial contour, and other structurally important regions. Evaluation without target-dataset fine-tuning examines whether the learned representation transfers beyond the occlusion patterns observed in the source training data. The corresponding cross-dataset comparison on the 68-point COFW test set is shown in (Fig. 7).

5.3.3. Evaluation under pose variations

We evaluate robustness to large pose changes on AFLW, reporting results on AFLW-Full and AFLW-Frontal following the standard protocol (Table 10). Among the listed methods, GCA-ODN obtains the lowest AFLW-Full error without Menpo pretraining. On AFLW-Frontal, the base GCA-ODN result is close to RHT-R, while GCA-ODN(M) improves to 1.01.

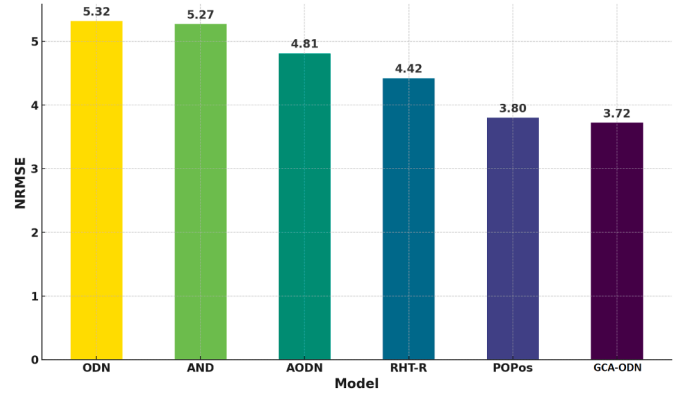


Fig. 7. Cross-dataset comparison on the 68-point COFW test set. GCA-ODN is evaluated without COFW fine-tuning.

Table 10

Comparison on AFLW-Full and AFLW-Frontal under the 19-landmark protocol (NRMSE $\times 10^{-2}$).

Method	Year	Full	Frontal
DCSD (Hannane et al., 2020)	2020	1.62	–
RetinaFace (Deng et al., 2020)	2020	1.41	1.11
AODN (Sadiq & Shi, 2022)	2022	1.38	1.13
RHT-R (Wan et al., 2023)	2023	1.99	1.02
POPos (Xiang et al., 2025)	2025	1.43	–
GCA-ODN	2026	1.30	1.03
GCA-ODN(M)	2026	1.26	1.01

Table 11

Comparison on the three official 300VW test categories (NRMSE $\times 10^{-2}$). † CAD-Net uses a distinct spatial-temporal architecture and published training protocol.

Method	Year	Category 1	Category 2	Category 3
ADN (Sadiq et al., 2019)	2019	4.75	4.34	6.72
AODN (Sadiq & Shi, 2022)	2022	4.69	4.26	6.67
CAD-Net† (Sadiq et al., 2026b)	2026	4.12	3.82	6.28
GCA-ODN	2026	4.40	4.01	6.28
GCA-ODN(M)	2026	4.31	4.01	6.21

5.3.4. Evaluation on videos

We evaluate temporal robustness on 300VW. Both GCA-ODN variants use the official 50-video training split; GCA-ODN(M) additionally incorporates Menpo pretraining. Evaluation is performed on all 64 official test videos across the three categories.

Relative to AODN, the base GCA-ODN model reduces NRMSE by approximately 6.2%, 5.9%, and 5.8% on Categories 1–3, respectively. Menpo pretraining provides an additional improvement on Categories 1 and 3, while the Category 2 result remains unchanged. CAD-Net reports lower errors on Categories 1 and 2 under its separate spatial-temporal implementation, whereas GCA-ODN(M) reports the lowest Category 3 value among the listed results.

6. Performance on emotion recognition

Facial expression recognition (FER) is a core capability for affect-aware neural perception systems in social robotics, assistive human-computer interaction, and remote interaction settings. In unconstrained environments, FER remains challenging because expression cues vary across individuals and are often corrupted by occlusion (e.g., masks and hands), head-pose changes, and illumination variations. Since facial landmarks encode structured geometric changes that correlate with facial action units (AUs) (Ekman & Friesen, 1978), we investigate whether the shared representation learned by GCA-ODN improves FER robustness under partial observation.

Table 12

Datasets and evaluation roles used in the FER experiments. AffectNet provides the source-domain training data for GCA-ODN, whereas CK+, JAFFE, and ISED are used only for cross-dataset testing.

Dataset	Usage	Size	Characteristics
AffectNet (Mollahosseini et al., 2017)	train/validation	>1M collected images	approximately 450,000 manually annotated images
CK+ (Lucey et al., 2010)	cross-dataset test	593 sequences; 123 subjects	327 sequences with validated expression labels
JAFFE (Lyons et al., 1998)	cross-dataset test	213 images; 10 subjects	six basic expressions and neutral
ISED (Happy et al., 2015)	cross-dataset test	50 subjects	spontaneous expressions extracted from videos

6.1. Experimental details for facial expression recognition

We evaluate GCA-ODN against representative FER baselines and emphasize cross-dataset generalization. GCA-ODN is trained using AffectNet expression supervision and evaluated directly on CK+, JAFFE, and ISED without target-domain fine-tuning. The comparison table also contains literature-reported values obtained under the original protocols of the corresponding studies. These literature values are included for contextual comparison and are not relabeled as controlled AffectNet-to-target experiments.

6.1.1. Datasets and specifications

We use four public datasets: AffectNet (Mollahosseini et al., 2017), CK+ (Lucey et al., 2010), JAFFE (Lyons et al., 1998), and ISED (Happy et al., 2015). AffectNet is used for training; CK+, JAFFE, and ISED are used for cross-dataset testing. The usage, size, and main characteristics of the FER datasets are summarized in (Table 12).

6.1.2. Preprocessing, training, and objective

For GCA-ODN, faces are detected and aligned when ground-truth bounding boxes are unavailable, cropped, and resized to 224×224 . Training augmentation includes random cropping, horizontal flipping, and rotation, and inputs are normalized using ImageNet channel statistics. The ResNet-18 backbone of GCA-ODN is initialized with ImageNet-pretrained weights (Deng et al., 2009b). These implementation details describe the proposed model; literature baselines retain the preprocessing and training protocols reported in their respective source publications.

No images or labels from CK+, JAFFE, or ISED are used for GCA-ODN model selection, hyperparameter tuning, or fine-tuning. AffectNet samples with expression labels activate the FER loss, while samples carrying landmark annotations activate the FLD loss through the label-availability formulation in Section 4. This avoids assigning artificial annotations to samples for which a task label is unavailable.

GCA-ODN is trained end-to-end with a joint objective that combines landmark regression and emotion classification (Section 4). For clarity, the task-dependent part of the joint loss is expressed as

$$\mathcal{L}_{\text{task}} = \alpha \mathcal{L}_{\text{FLD}} + \beta \mathcal{L}_{\text{FER}}, \quad (32)$$

where $\mathcal{L}_{\text{FLD}}$ and $\mathcal{L}_{\text{FER}}$ follow Eqs. (23) and (24). We set $\alpha = 0.4$ and $\beta = 0.6$ based on validation performance (Table 4). Unless otherwise stated, we use the optimizer and schedule reported in Table 6. The regularization terms in Eq. (22) remain active during joint optimization and are omitted from Eq. (32) only to simplify the notation. The FER experiments retain the fixed GCA-ODN module settings $\tau_{\text{adm}} = 0.5$, $p = 0.5$, $r_{\text{gc}} = 8$, and $r_A = 8$.

6.2. Cross-dataset evaluation results

Table 13 reports accuracy on CK+, JAFFE, and ISED. GCA-ODN and CAD-Net are trained using AffectNet expression supervision and evaluated on the target datasets without fine-tuning. The other entries are literature-reported comparison values whose original training and evaluation protocols are not necessarily identical. This distinction prevents dataset-specific or in-domain results from being incorrectly presented as controlled cross-dataset measurements.

GCA-ODN obtains 84.6%, 62.1%, and 68.3% on CK+, JAFFE, and ISED, respectively. These values exceed the listed literature-reported CNN, domain-adaptation, and attention-based results wherever a corresponding target value is available. Because the original baseline protocols vary, these differences are interpreted as contextual rather than strictly controlled effect sizes.

CAD-Net reports higher accuracies of 87.3%, 65.1%, and 70.2% under its published AffectNet-to-target setting. CAD-Net uses recurrent criss-cross geometry modeling, an attentive dropout block, and a low-rank landmark-regression formulation that differ from the Global Context Module and shared low-rank refinement used by GCA-ODN (Sadiq et al., 2026b). Its results are therefore presented transparently as those of a separate architecture and are not transferred to GCA-ODN.

Recent transformer-based FER approaches, including MMATrans, POSTER++, and related 2024–2025 methods, are commonly evaluated on RAF-DB, AffectNet, FERPlus, or dedicated occluded-FER benchmarks rather than under the AffectNet-to-CK+/JAFFE/ISED protocol (Liu et al., 2024a; Wang et al., 2026). Their reported values are therefore discussed in the related-work section but are not inserted into Table 13 as though they were directly comparable CK+, JAFFE, or ISED transfer results. This protocol-aware treatment addresses recent transformer developments without fabricating missing target-dataset measurements.

6.3. Geometry-aware representation and FER robustness

Many FER models rely heavily on appearance cues and may overfit to dataset-specific textures or backgrounds. In contrast, GCA-ODN incorporates explicit geometric supervision through FLD, which is intended to encourage the shared embedding to remain stable under occlusion and nuisance variations. Global context modeling supports holistic structural consistency, attentive dropout reduces spurious reliance on dominant cues, and low-rank refinement promotes compact representations. Together, these mechanisms contribute to the cross-dataset results in Table 13.

Table 14 provides an incremental comparison of three GCA-ODN configurations trained using the same backbone and preprocessing. The first configuration uses conventional FER-only training. The second uses the GCA-ODN representation without ADM and without joint landmark supervision. The final configuration uses the complete joint FLD-FER model.

Relative to the FER-only backbone, the complete GCA-ODN configuration improves accuracy by 6.2, 1.6, and 4.7 percentage points on CK+, JAFFE, and ISED, respectively. Relative to the FER-only GCA-ODN configuration without ADM, the corresponding gains are 2.2, 0.5, and 2.8 percentage points.

These results support the complete geometry-aware design but do not independently establish the contribution of ADM or joint landmark supervision, because both factors change between the second and third configurations. A causal attribution to either component would require additional variants such as joint FLD-FER training without ADM and FER-only training with ADM enabled. Such unmeasured values are not inferred from CAD-Net or from its FER-specific ablations.

The reported target datasets provide a demanding cross-dataset transfer setting, but they are not dedicated occlusion-severity benchmarks. Consequently, Table 13 supports cross-dataset generalization under heterogeneous capture conditions rather than a controlled claim

Table 13

FER accuracy (%) on CK + , JAFFE, and ISED. Literature-reported baselines follow their original protocols and are included for context. GCA-ODN and CAD-Net use AffectNet-to-target evaluation without target-domain fine-tuning.

Method	Year	Evaluation status	CK +	JAFFE	ISED
CNN (Kennedy & Balint, 2016)	2016	literature reported	72.8	50.2	59.3
CNN (Xia et al., 2017)	2017	literature reported	62.8	48.4	51.6
EmotionalDAN (Tautkutė & Trzciński, 2019)	2019	literature reported	73.5	46.5	62.0
AGRA (Xie et al., 2020)	2020	literature reported	77.5	61.0	–
AdaFER (Peng et al., 2022)	2022	literature reported	81.4	61.4	–
PACVT (Liu et al., 2023a)	2023	literature reported	82.1	–	–
GCA-ODN	2026	AffectNet→target; no FT	84.6	62.1	68.3
CAD-Net (Sadiq et al., 2026b)	2026	AffectNet→target; no FT	87.3	65.1	70.2

Table 14

Incremental comparison of GCA-ODN FER configurations under AffectNet-to-target evaluation. The second and third rows differ in both ADM usage and task supervision; the comparison therefore does not isolate either factor independently.

Variant	CK +	JAFFE	ISED
Backbone (FER-only)	78.4	60.5	63.6
GCA-ODN (FER-only; without ADM)	82.4	61.6	65.5
GCA-ODN (joint FLD + FER)	84.6	62.1	68.3

across known occlusion ratios. Recent work using explicit occlusion detection, FaceOcc-based synthesis, and dedicated datasets such as Occlusion-RAF-DB and FED-RO provides a complementary evaluation setting (Wang et al., 2026). GCA-ODN does not use that detector–masking–recovery pipeline, and its results on those datasets are not claimed here.

All GCA-ODN values in this section are reported as the available experimental measurements. No mean–standard-deviation values, paired significance tests, or confidence intervals are claimed without repeated GCA-ODN runs and per-sample prediction records. Literature-only results are likewise excluded from paired statistical testing because their prediction outputs are unavailable.

7. Ablation analysis

This section evaluates the contribution of each component in GCA-ODN: the **Global Context Module (GCM)**, the **Attentive Dropout Module (ADM)**, and the **Low-Rank Feature Refinement Module (LM)**. The objective is to assess whether the observed performance gains are associated with complementary mechanisms—global structure preservation, attention-guided suppression of dominant or potentially unreliable responses, and compact refinement—rather than with a single architectural change. The analysis includes a leave-one-module-out FLD study and an incremental FER comparison. The conclusions are restricted to experimentally verified configurations; unmeasured single-module, ADM sub-component, interaction, or sensitivity results are not inferred from the available observations.

7.1. Leave-one-module-out ablation on 300W-challenging

We conduct a controlled ablation study on the 300W Challenging subset, which contains severe occlusion and large pose variations. All variants use the same backbone (ResNet-18), input resolution (224×224), data augmentation, and training schedule to ensure a consistent comparison protocol. Table 15 reports the results in terms of NRMSE ($\times 10^{-2}$).

The pairwise configurations are interpreted as leave-one-module-out variants of the complete architecture: GCM + LM removes ADM, GCM + ADM removes LM, and ADM + LM removes GCM. This formulation measures the change caused by removing each module while retaining the other two components.

As shown in Table 15, every leave-one-module-out configuration performs worse than the complete model. The full GCM + ADM + LM configuration reduces NRMSE by 1.60×10^{-2} relative to the ResNet-18 baseline, corresponding to a relative error reduction of 22.19%.

Removing ADM produces the largest leave-one-out degradation, increasing NRMSE from 5.61 to 6.90. Removing LM increases the error to 6.70, whereas removing GCM increases it to 6.60. Within this controlled setting, the results indicate that none of the three modules is redundant. ADM has the largest removal effect, while GCM and LM also provide measurable complementary contributions. Because the modules interact within the fused representation, these differences should be interpreted as conditional leave-one-out effects rather than as independent or additive estimates of isolated module performance.

The available table does not include configurations containing only GCM, only ADM, or only LM. It therefore establishes the importance of each component within the complete architecture but does not quantify the standalone performance of each module. No missing single-module values are inferred from CAD-Net, ADODN, or another related architecture.

7.2. FER-oriented ablation

The effect of the complete architecture on FER is examined using the incremental configurations reported in Table 14. All variants use the same backbone, preprocessing pipeline, AffectNet source-domain training data, and cross-dataset evaluation targets.

Relative to the FER-only ResNet-18 backbone, the complete joint model improves accuracy by 6.2 percentage points on CK + , 1.6 points on JAFFE, and 4.7 points on ISED. Relative to the FER-only GCA-ODN configuration without ADM, the corresponding improvements are 2.2, 0.5, and 2.8 percentage points.

These results support the complete geometry-aware architecture for cross-dataset FER. However, the comparison between the second and third configurations changes both ADM usage and task supervision. It therefore demonstrates the combined benefit of the complete joint design but does not isolate ADM from FLD supervision. A fully controlled factorial comparison would additionally require an FER-only model with ADM enabled and a joint FLD–FER model with ADM disabled. These measurements are not available and are not inferred from the present results.

The FER values are reported as the available point estimates. Mean–standard-deviation values, confidence intervals, and statistical significance are not claimed without repeated GCA-ODN runs and per-sample prediction records.

7.3. Hyperparameter scope

The Global Context reduction ratio was examined over $r_{gc} \in \{8, 16, 32, 64\}$, and $r_{gc} = 8$ is used in the reported configuration. The remaining module settings are fixed at $\tau_{adm} = 0.5$, $p = 0.5$, and $r_A = 8$, with deterministic importance-map weighting used during inference, as specified in Section 5.2.

Table 15

Leave-one-module-out ablation on 300W Challenging. The reduction column is calculated relative to the ResNet-18 baseline, whereas degradation relative to the complete model is reported as the absolute increase in NRMSE. NRMSE values are reported as $\times 10^{-2}$.

Configuration	GCM	ADM	LM	NRMSE	Reduction vs. baseline (%)	Increase vs. full
ResNet-18	–	–	–	7.21	0.00	1.60
Without ADM	✓	–	✓	6.90	4.30	1.29
Without LM	✓	✓	–	6.70	7.07	1.09
Without GCM	–	✓	✓	6.60	8.46	0.99
Complete GCA-ODN	✓	✓	✓	5.61	22.19	0.00

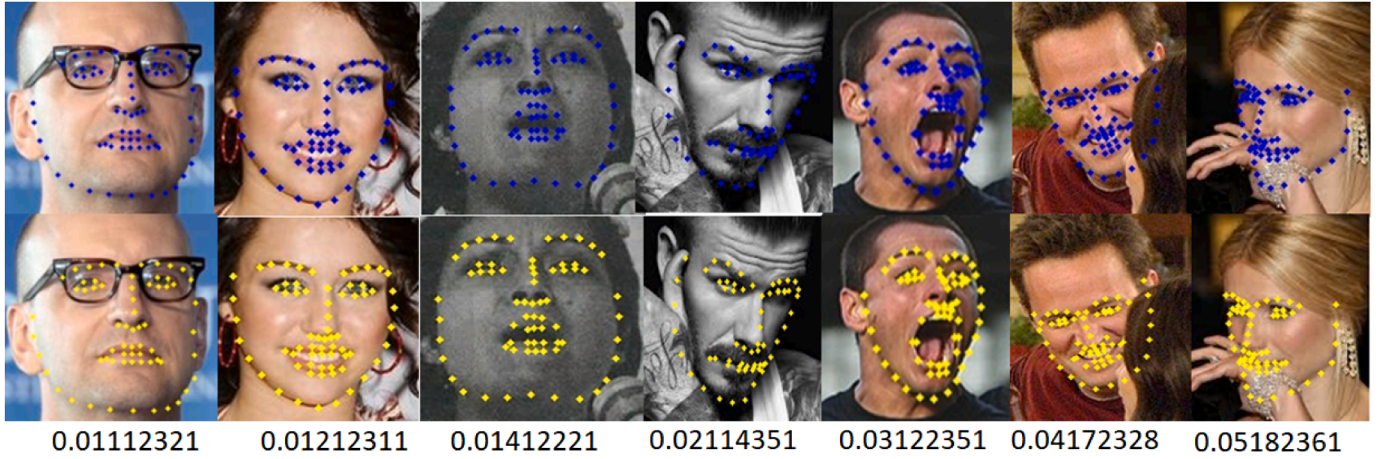


Fig. 8. Qualitative facial landmark localization examples on 300W Challenging. Ground-truth landmarks are shown in blue in the upper row, and GCA-ODN predictions are shown in yellow in the lower row. The examples include partial occlusion, pose variation, and difficult appearance conditions.

The available GCA-ODN measurements do not provide complete numerical sensitivity curves for τ_{adm} , p , or r_A . Accordingly, this section reports their selected values but does not introduce unmeasured NRMSE or FER accuracy results. Sensitivity findings from CAD-Net or other attentive-dropout and low-rank models are not transferred because the corresponding modules and optimization procedures differ. The quantitative ablation conclusions are therefore restricted to Tables 14 and 15.

7.4. Qualitative behavior and limitations

Fig. 8 provides qualitative examples under challenging conditions. Ground-truth landmarks are shown in blue, and predictions are shown in yellow. The examples illustrate representative cases in which GCA-ODN maintains a plausible facial configuration despite partial loss of local evidence. They are intended to complement, rather than replace, the quantitative benchmark results.

The model remains vulnerable when several important geometric anchors are simultaneously unavailable, such as simultaneous coverage of the eye and mouth regions, extreme out-of-plane rotation, very low spatial resolution, strong motion blur, or inaccurate face cropping. Under these conditions, global context and low-rank correlation cannot reconstruct facial evidence that is absent or severely corrupted across all visible regions.

The present qualitative analysis does not quantify failure frequency or performance as a function of controlled occlusion severity. A dedicated failure-case study should therefore include representative unsuccessful predictions together with controlled synthetic occlusions at increasing area ratios. Such measurements are identified as future experimental work rather than presented here without supporting results.

8. Applications

The following applications represent potential research and deployment directions rather than systems clinically, legally, or operationally

validated in this study. Robust facial landmark detection and expression recognition under partial observation can support a range of neural perception systems. (Fig. 9) illustrates representative potential application scenarios for occlusion-robust facial landmark detection and expression recognition.

In digital identity and access control, reliable landmark localization can improve face alignment and feature normalization, strengthening biometric pipelines against nuisance factors such as pose variation, illumination changes, and partial occlusion. Stable landmark trajectories may also complement liveness cues derived from facial dynamics, such as blinking or short-term landmark motion. However, GCA-ODN is not evaluated here as a complete anti-spoofing, identity-verification, or biometric-authentication system, and its landmark predictions alone should not be interpreted as evidence of presentation-attack resistance.

In human–computer interaction and cyber-physical systems, accurate landmarks can support gaze estimation, facial gesture interfaces, and behavioral analytics. Landmark-derived head orientation and eye-region cues may be useful for attention-aware interfaces, driver monitoring, and accessibility systems. These downstream tasks require task-specific output heads, annotations, evaluation metrics, and safety testing beyond the FLD and FER experiments presented here.

Facial landmark analysis further supports affective computing by providing a structured representation for FER. Landmark-aware emotion inference may assist affect-aware remote interaction and adaptive interfaces under acquisition constraints such as motion blur and partial occlusion. Emotion predictions should nevertheless be treated as uncertain behavioral signals rather than direct measurements of psychological state, stress, anxiety, intention, or clinical condition.

In clinical and biomedical scenarios, face masks, hair, hands, and instruments can hide facial regions and reduce the reliability of conventional appearance-based models. The proposed combination of global context aggregation, attention-guided feature suppression, and low-rank refinement may support research on remote patient observation, rehabilitation interfaces, pain-related expression analysis, and assistive care.

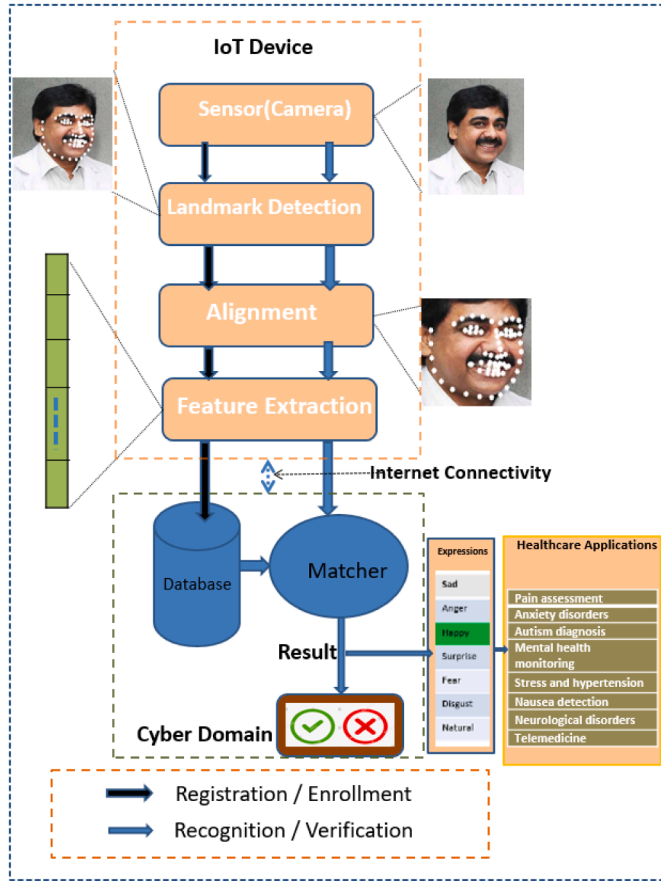


Fig. 9. Potential research applications of occlusion-robust facial landmark detection and expression recognition. The illustrated applications require task-specific datasets, evaluation, and validation beyond the benchmark experiments reported in this paper.

The present experiments use public computer-vision benchmarks and do not constitute clinical validation, diagnostic evidence, treatment evaluation, or testing on a patient cohort.

The shared representation may also be extended to head-pose estimation by attaching a task-specific regression head for yaw, pitch, and roll. GCM could provide holistic orientation cues, while ADM could reduce dependence on individual pose-sensitive regions. Because no head-pose dataset or angular-error evaluation is included in this study, head-pose estimation remains an architectural extension rather than an experimentally validated application.

Practical deployment would additionally require evaluation of demographic performance differences, calibration, privacy protection, informed consent, data governance, failure detection, and appropriate human oversight. These considerations are particularly important in healthcare, biometrics, education, employment, and other high-impact settings.

9. Conclusion and future work

This paper presented **GCA-ODN**, a computationally practical deep architecture for joint facial landmark detection and facial expression recognition under occlusion, pose variation, and illumination changes. GCA-ODN integrates three complementary components: (i) a Global Context Module to preserve long-range structural dependencies, (ii) an Attentive Dropout Module to reduce over-reliance on dominant activations through attention-guided masking, and (iii) a Low-Rank Feature Refinement Module to promote compact, noise-suppressed shared embeddings.

Across the evaluated protocols, extensive evaluations demonstrate consistent improvements in accuracy and robustness. On 300W, the base model obtains NRMSE values of 2.60×10^{-2} , 3.20×10^{-2} , and 5.61×10^{-2} on the Common, Full, and Challenging subsets, respectively. In AffectNet-to-target cross-dataset FER evaluation without target-domain fine-tuning, it reaches 84.6%, 62.1%, and 68.3% on CK+, JAFFE, and ISED. The leave-one-module-out analysis shows that removing GCM, ADM, or LM increases the 300W-Challenging error, supporting their complementary roles within the complete architecture.

The CNN-based design does not universally outperform every recent transformer or geometry-aware model. Its principal contribution is the integration of local convolutional features, lightweight global-context aggregation, attentive feature regularization, and low-rank refinement within a joint FLD-FER framework that processes more than 80 frames per second on 224×224 inputs with batch size one on an NVIDIA RTX A6000. Results obtained by external methods using different architectures, additional facial data, multi-dataset training, or target-domain optimization are therefore treated as contextual rather than strictly controlled comparisons.

Several limitations remain. The current evidence does not include a complete profiler-derived comparison of parameters, FLOPs, peak memory, and latency across the final GCA-ODN implementation and representative baselines. Repeated-run variability, confidence intervals, and paired statistical tests are also unavailable. The ablation study quantifies leave-one-module-out effects but does not provide standalone single-module experiments, a complete ADM drop-mask versus importance-map factorial analysis, or an isolated comparison between joint supervision and ADM. Complete numerical sensitivity curves for r_{adm} , p , and r_A are not available.

The present evaluation also lacks controlled synthetic-occlusion results over increasing severity levels and a quantitative failure-rate analysis for simultaneous multi-region occlusion, severe blur, low resolution, and inaccurate face detection. Finally, the proposed extensions to clinical monitoring, biometric security, and head-pose estimation have not been directly validated in this study.

Future work will therefore focus on profiler-derived complexity evaluation, repeated-run statistical analysis, fine-grained module and task-supervision ablations, complete hyperparameter sensitivity studies, and controlled synthetic occlusion across increasing severity levels. We will also investigate lightweight transformer and CNN-transformer hybrid backbones under an identical training protocol, uncertainty-aware prediction, temporal facial modeling, head-pose estimation, 3D facial analysis, and multimodal affect inference. Clinical, biometric, or security-related extensions will require domain-specific datasets, calibration and fairness analysis, ethical review, privacy safeguards, and prospective validation before operational use.

CRediT authorship contribution statement

Muhammad Sadiq: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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